AI-Based Occlusion Handling in VR Using Monocular Depth and Segmentation Models

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Introduction to Problem

← Outline

Why Occlusion Matters in AR/VR

- Immersive AR/VR experiences depend on accurate spatial interaction between virtual and real-world objects.
- When a virtual object fails to hide behind a real one, the illusion of depth collapses and the scene feels visually inconsistent.
- Occlusion reasoning — deciding which object should be visible at each pixel — is therefore a core requirement for believable mixed reality.

The Core Question

"Should a virtual object be hidden behind a real-world object in the current frame?"

Answering this reliably in real time requires per-pixel depth and a clean foreground mask of the real scene.

Introduction to Problem

← Outline

Current Industry Approach and Its Limitations

Hardware-Dependent Sensors

Commercial AR/VR systems rely on LiDAR, Time-of-Flight, and structured-light depth sensors to capture scene geometry.

Cost and Power Overhead

Dedicated depth hardware increases device price, battery drain, and overall system complexity for headsets and phones.

Limited Accessibility

Most low-cost devices and standard webcams lack these sensors, leaving a large class of hardware unable to support occlusion-aware AR.

Motivation: Can accurate occlusion be achieved using only a standard RGB camera, removing the dependency on specialised depth hardware?

Existing Research Overview

← Outline

Two Foundational Directions in the Literature

Monocular Depth Estimation

- MiDaS (Ranftl et al., 2020) demonstrated robust zero-shot depth estimation by mixing multiple datasets.
- Enables per-pixel relative depth from a single RGB frame, with no stereo or LiDAR.
- Forms the depth backbone used in this work through an ONNX deployment.

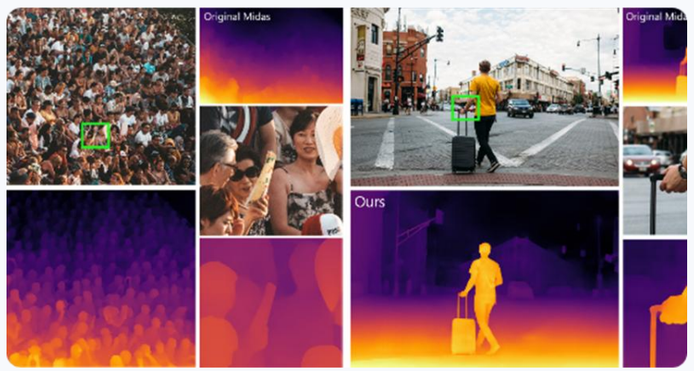


Fig. Example MiDaS depth output (plasma colormap)

Semantic / Salient Segmentation

- U_2-Net (Qin et al., 2020) introduced a nested U-structure for strong salient object detection at low cost.
- Produces clean binary foreground masks that isolate real-world objects from the background.
- Used here to identify real pixels whose depth must be compared against the virtual object.

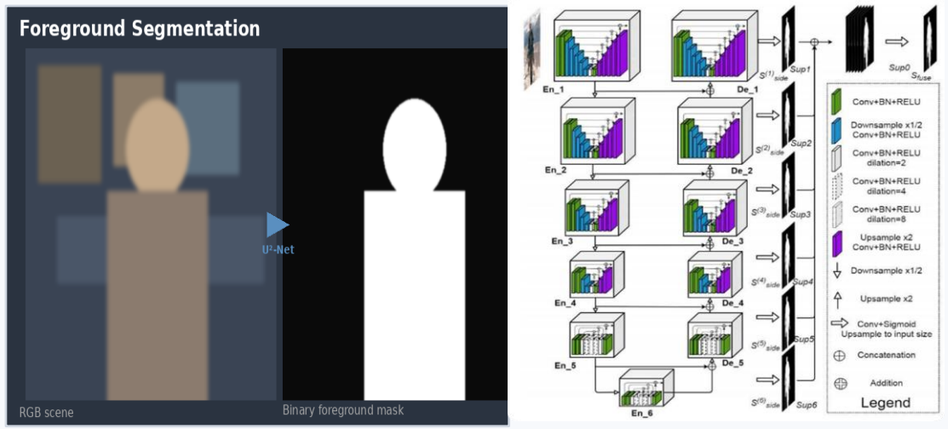


Fig. RGB scene → U²-Net binary foreground mask

Existing Research Overview

← Outline

Commercial AR Platforms

Google ARCore Depth API

Google's ARCore exposes a Depth API for object occlusion in AR, but relies on supported devices and motion-based depth, limiting static single-frame and webcam-only use.

Apple ARKit with LiDAR

On recent iPhones and iPads, ARKit uses the built-in LiDAR scanner to produce dense depth maps, achieving high-quality occlusion at the cost of premium, sensor-equipped hardware.

What the Literature Tells Us

- Strong monocular depth models now exist and are accessible.
- Robust, lightweight segmentation models are readily available.
- Commercial AR occlusion remains tied to motion cues or hardware sensors.
- **A purely monocular, sensor-free occlusion pipeline that combines both models is not widely demonstrated.**

Research Gaps and Objectives

← Outline

Identified Research Gaps

1 Hardware Dependency

Most working occlusion pipelines assume a depth sensor, excluding standard laptops, webcams, and low-cost headsets.

2 Fragmented Use of Models

Monocular depth and segmentation are widely studied in isolation, but rarely combined into a single occlusion reasoning pipeline.

3 Limited XR Headset Validation

Few works demonstrate the feasibility of a purely software-based occlusion pipeline on standalone XR headsets.



Research Gaps and Objectives

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Research Objectives

Design a Hardware-Free Framework

Build an occlusion-handling pipeline that runs on monocular RGB input alone, with no dedicated depth sensor.

Integrate Depth + Segmentation

Combine a MiDaS-based depth model with a U²-Net-based segmentation model inside a single reasoning module.

Implement a Working Prototype

Deploy the pipeline in Python using OpenCV and ONNX models, validated on a standard laptop webcam.

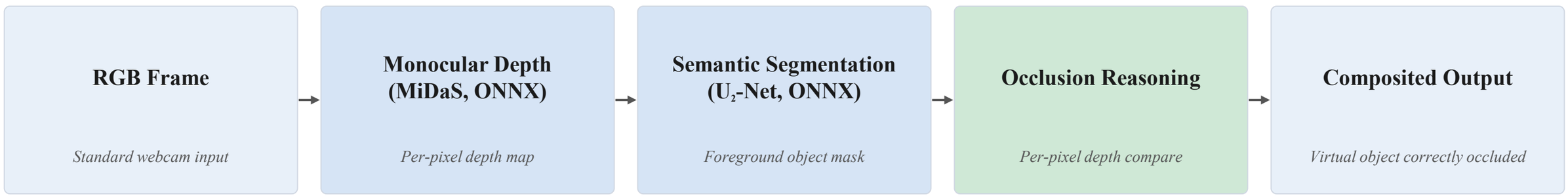
Demonstrate XR Readiness

Evaluate feasibility for deployment on standalone AR/VR headsets, with planned integration on the AJNAXR headset.

Research Approach and Methodology

← Outline

System Overview — Three-Stage Pipeline



1. **Monocular Depth Estimation —** a MiDaS-based ONNX model estimates pixel-level depth from each RGB frame.
2. **Semantic Segmentation —** a U_2-Net-based ONNX model produces a binary foreground mask to isolate real-world objects.
3. **Occlusion Reasoning Module —** compares the real-world object's depth against the assigned depth of the virtual object; if the real pixel is closer, the virtual pixel is hidden.

Research Approach and Methodology

← Outline

Occlusion Decision — Per-Pixel Depth Comparison

Governing Rule

$$\mathbf{O(x, y)} = \begin{cases} I(x, y) & \textit{if } D(x, y) < D_v \\ V(x, y) & \textit{otherwise} \end{cases}$$

Where:

I(x, y) — real-world image pixel at (x, y).
V(x, y) — virtual object pixel at (x, y).
D(x, y) — estimated depth of the real scene.
D_v — assigned depth of the virtual object.

Implementation Details

Language: Python
Vision / IO: OpenCV
Inference: ONNX Runtime (MiDaS + U_2-Net)
Input: Standard laptop webcam (RGB only)
Processing: CPU-only, no GPU or depth sensor required
Output: Composited frame with occluded virtual object
Target deployment: Standalone AR/VR headsets (AJNAXR planned)

Target deployment device:



Proposed Work

← Outline

A Hardware-Free Occlusion Reasoning Framework

Core Contribution

- A monocular, AI-driven occlusion pipeline that replaces LiDAR and Time-of-Flight sensors with standard RGB input.
- Unified module that fuses per-pixel depth and foreground segmentation into a single occlusion decision per frame.
- Full Python + OpenCV + ONNX implementation validated on a standard laptop webcam with no auxiliary hardware.
- Designed from the outset for deployment on standalone AR/VR headsets, with planned integration on the AJNAXR headset.

Why It Matters

Cost-effective: removes the need for dedicated depth hardware.

Scalable: runs on commodity laptops and potentially on XR headsets without specialised sensors.

Adaptable: modular pipeline where individual models can be swapped for lighter or more accurate alternatives.

Accessible: opens occlusion-aware AR to a much wider class of devices.

Results and Analysis

← Outline

Experimental Demonstration

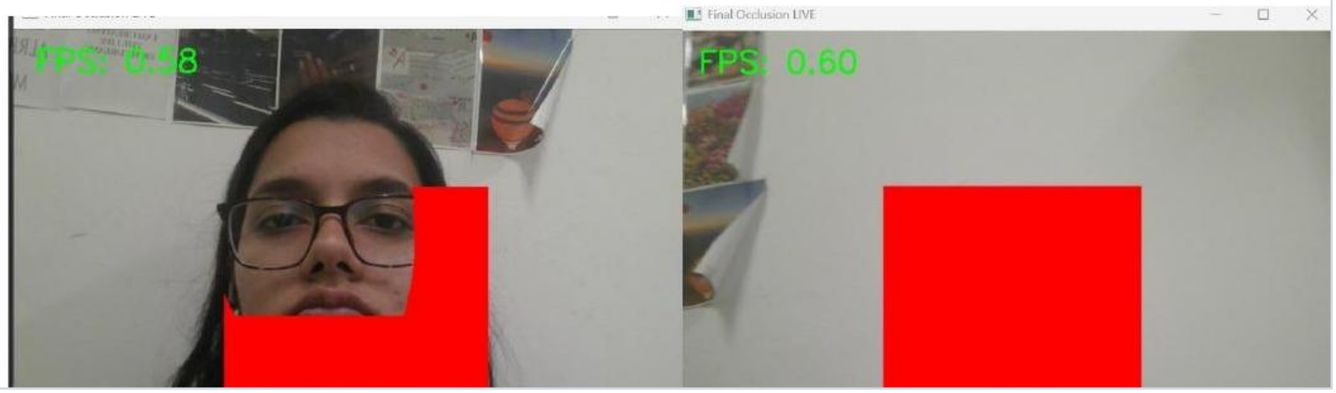


Fig. 1. Left: virtual object correctly hidden behind the foreground subject. Right: virtual object fully rendered in the absence of foreground objects.

0.5 – 1
FPS on CPU-only laptop

0 sensors
Runs on standard RGB webcam

Key Observations

Correct Occlusion Behaviour

Virtual object is properly hidden when the real subject is closer to the camera.

Runtime Performance

0.5–1 FPS on a CPU-only setup using a standard laptop webcam.

Hardware-Free Operation

No LiDAR, ToF, or depth sensor required — only an RGB camera.

Conclusion and Future Work

← Outline

Conclusion

- Hardware-independent occlusion handling is feasible using only monocular vision-based AI models.
- The prototype reproduces realistic occlusion effects without any dedicated depth hardware.
- Results validate the effectiveness of AI-based models for efficient per-pixel depth reasoning.
- Though throughput is currently limited by model size and CPU-only inference, the framework is efficient enough to support XR devices.

Future Work

1. **Real-Time Performance**
   Optimise inference pipeline to raise frame rate beyond 1 FPS.
2. **Lightweight Model Assessment**
   Evaluate compressed and distilled variants of MiDaS and U²-Net.
3. **XR Headset Integration**
   Deploy and validate on the AJNAXR standalone headset.
4. **Robustness Evaluation**
   Test across varied lighting, backgrounds, and scene complexity.

References

← Outline

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